

Tema 2

Ejercicio N°1

$$r = 0,02 \text{ dm}$$

$$r = 2 \text{ dm}$$

$$\text{Vol. esf} = \frac{4}{3}\pi r^3 = \frac{4}{3} \cdot \pi \cdot 2^3 \text{ dm}^3 = 33,51 \text{ dm}^3 = \underline{\underline{33,51 \text{ l}}}$$

Ejercicio N°2

$$\vec{A} = (8,0\hat{i} - 5,4\hat{j} + 6,5\hat{k})$$

$$\vec{B} = (-3\hat{i} + 10\hat{j} + 4,7\hat{k})$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8,0 & -5,4 & 6,5 \\ -3 & 10 & 4,5 \end{vmatrix} = -90,38\hat{i} - 57,1\hat{j} + 63,8\hat{k}$$

$$|\vec{A} \times \vec{B}| = \sqrt{(-90,38)^2 + (-57,1)^2 + (63,8)^2} =$$

$$A_{\text{rez}} = 124,49 \text{ cm}^2 \sim \underline{\underline{124 \text{ cm}^2}}$$

Ejercicio N°3

pero determine la aceleración

$$x = v_0 t + \frac{1}{2} a t^2$$

$$a = \frac{2x}{t^2} = \frac{2 \cdot 9 \text{ m}}{3^2 \text{ s}^2} = 2 \text{ m/s}^2$$

$$v_{fx} = v_{ix} + at$$

$$t = \frac{v_{fx} - v_{ix}}{a} = \frac{24 \text{ m/s} - 0}{2 \text{ m/s}^2} = \underline{\underline{12 \text{ s}}}$$

Ej N°4

$$y = v_{iy} \cdot t + \frac{1}{2} g_y \cdot t^2$$

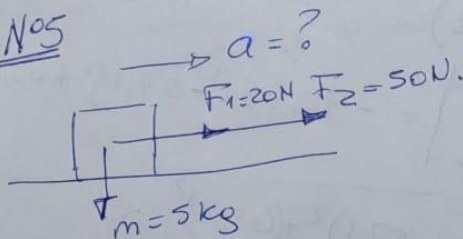
$$-100 \text{ m} = 0 + \frac{1}{2} \cdot (-9.8) \frac{\text{m}}{\text{s}^2} \cdot t^2$$

$$t = \sqrt{\frac{-100 \text{ m} \times 2}{-9.8 \text{ m/s}^2}} = 4.52 \text{ s.}$$

luego aplico $x = v_{ix} \cdot t = 15 \text{ m/s} \times 4.52 \text{ s}$

$$\underline{\underline{x = 67.8 \text{ m}}}$$

Ej N°5



$\rightarrow a = ?$

$$\sum F_x = m \cdot a$$

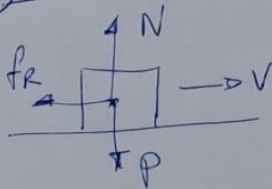
$$F_1 + F_2 = m \cdot a$$

$$20 \text{ N} + 50 \text{ N} = 5 \text{ kg} \cdot a$$

$$\frac{70 \text{ N}}{5 \text{ kg}} = a$$

$$\underline{\underline{a = 14 \text{ m/s}^2}}$$

Ej N°6



$$W = \Delta E_c$$

si $v = 0$ entonces $E_c = 0$

$$W = E_{c_f} - E_{c_0} =$$

$$W = 0 - \frac{1}{2} m \cdot v^2 = -\frac{1}{2} \cdot 1 \text{ kg} \cdot \frac{2 \text{ m}}{\text{s}^2} =$$

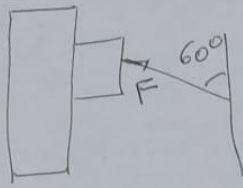
$$W = -2 \text{ J.}$$

Luego.

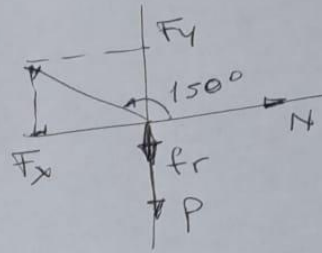
$$W = F_R \cdot d \cdot \cos 180^\circ$$

$$W = -2 \text{ J} \text{ entonces } F_R = \frac{-2 \text{ J}}{5 \text{ m} \cdot (-1)} = \underline{\underline{0.4 \text{ N}}}$$

Ejercicio N° 8



Si el cuerpo se mueve hacia arriba la fr va hacia abajo.



$$\sum F_x = 0$$

$$N - F \cdot \cos 150^\circ = 0$$

$$N = F \cdot \cos 150^\circ$$

$$\sum F_y = -fr + F \cdot \sin 150^\circ - P$$

$$-\mu \cdot F \cdot \cos 150^\circ + F \cdot \sin 150^\circ = P$$

$$F(-\mu \cdot \cos 150^\circ + \sin 150^\circ) = P$$

$$F = \frac{P}{(-\mu \cdot \cos 150^\circ + \sin 150^\circ)} = \frac{1200 \text{ kg}}{(-0,25 \cdot \cos 150^\circ + \sin 150^\circ)}$$

$$\boxed{F = 1675 \text{ N}}$$

Ejercicio N° 7

$$h = 100 \text{ m.}$$

$$g = 9,8 \text{ m/s}^2$$

$$v_f = 0 \text{ m/s}$$

$$v_0 = ?$$

$$v_f^2 = v_i^2 - 2g \cdot h$$

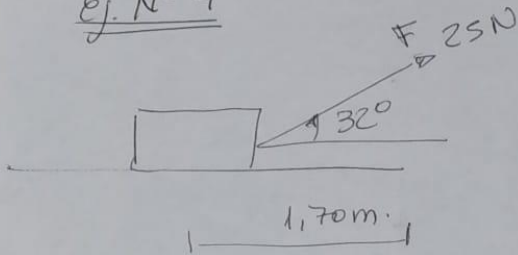
$$v_f^2 = 0 \text{ m/s}^2$$

$$0 = v_i^2 - 2 \cdot 9,8 \frac{\text{m}}{\text{s}^2} \cdot 100 \text{ m}$$

$$v_i = \sqrt{2 \cdot 9,8 \frac{\text{m}}{\text{s}^2} \cdot 100 \text{ m}}$$

$$\boxed{v_i = 44,27 \text{ m/s}}$$

Ej. N° 9

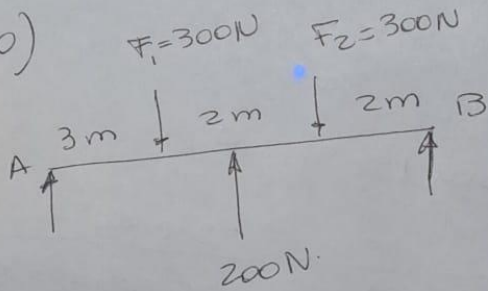


$$W = F \cdot d \cdot \cos 32^\circ$$

$$W = 25 \text{ N} \times 1.70 \text{ m} \times \cos 32^\circ$$

$$W = 36.04 \text{ J}$$

10)



$$\sum F_y = 0$$

$$T_A - F_1 - F_2 + 200 \text{ N} + T_B = 0$$

$$\sum M_A = 0$$

$$-F_1 \cdot 3 \text{ m} + 200 \cdot 3.5 \text{ m} - F_2 \cdot 5 \text{ m} + T_B \cdot 7 \text{ m} = 0$$

$$-300 \text{ N} \cdot 3 \text{ m} + 200 \text{ N} \cdot 3.5 \text{ m} - 300 \text{ N} \cdot 5 \text{ m} = -T_B \cdot 7 \text{ m}$$

$$-1700 \text{ N} \cdot \text{m} = -T_B \cdot 7 \text{ m}$$

$$T_B = 242.85 \text{ N}$$

luego Reemplazo en $\sum F_y = 0$

$$T_A - 300 \text{ N} - 300 \text{ N} + 200 \text{ N} + 242.85 \text{ N} = 0$$

$$T_A = 157.15 \text{ N}$$